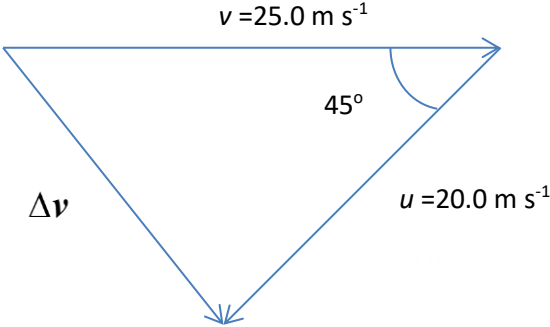


1	(a)	unit of D = (units of k)(units of d) units of k = 1 unit of Q = (units of C)(units of ρ)(units of $\sqrt{g}$)(units of D) ^{3/2} $\text{kg s}^{-1} = (\text{units of C})(\text{kg m}^{-3})(\sqrt{\text{m s}^{-2}})(\text{m}^{3/2})$ units of C = m	C1 A1 A1
	(b)	$\rho = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi r^3} = \frac{3m}{4\pi r^3}$ $r^3 = \frac{3m}{4\pi\rho} = \frac{3(500)}{4\pi(7.75)} = 15.4$ $r = 2.4879 \text{ cm}$ $\frac{\Delta r}{r} = \frac{1}{3} \left(\frac{\Delta m}{m} + \frac{\Delta \rho}{\rho} \right) = \frac{1}{3} \left(\frac{1}{500} + \frac{0.01}{7.75} \right) = \frac{1}{3} (0.00329)$ $\Delta r = 0.003 \text{ cm}$ Hence $r = 2.488 \pm 0.003 \text{ cm}$	C1 C1 A1
	(c)	$u = 20.0 \text{ m s}^{-1}$ in north-east direction, $v = 25.0 \text{ m s}^{-1}$ in east direction $\Delta v = v - u = \Delta v = v + (-u)$  Using the cosine rule, $\Delta v = \sqrt{v^2 + u^2 - 2vu \cos 45^\circ}$ $= \sqrt{25^2 + 20^2 - 2(25)(20) \cos 45^\circ}$ Hence $\Delta v \approx 17.830 \text{ m s}^{-1}$ $F = ma = m \frac{\Delta v}{\Delta t} = 800 \left(\frac{17.8}{3.0} \right) = 4747 \text{ N} \approx 4.75 \times 10^3 \text{ N}$	C1 M1 A1